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An unsupervised domain adaptation deep learning method for spatial and temporal transferable crop type mapping using Sentinel-2 imagery

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Abstract

Accurate crop type mapping is essential for crop growth monitoring and yield estimation. Recently, various machine learning methods have been increasingly used for crop type mapping, but they often lose their validity when directly applied to other regions and years due to differences in the distribution of source and target data, that is, domain shift. To address the problem, we developed a deep adaptation crop classification network (DACCN) based on the idea of unsupervised domain adaptation (UDA). The proposed DACCN mainly consists of two parts, a feature extractor that converts the original input into high-level representations, and a domain aligner where the discrepancy between source and target distributions is measured using multiple kernel variant of maximum mean discrepancy (MK-MMD). Four states in the United States (U.S.) Corn Belt and two provinces in northeastern China were used as study areas, where samples used for model building and accuracy evaluation were collected based on time-series Sentinel-2 imagery and reference maps in 2018 and 2019. Then, three experiments were designed to verify the transferability of DACCN across space, time, and space–time, respectively. In each experiment, the proposed DACCN was compared to deep crop classification network (DCCN), a model with a similar structure to DACCN but without the domain adaptation mechanism, and two machine learning methods, random forest (RF) and support vector machines (SVM). The experimental results showed that DACCN outperformed other models in most transfer cases with overall classification accuracies ranging from 0.835 to 0.922. The DACCN also performed better in spatially continuous mapping with its predicted crop type maps more consistent with the reference ones. As an innovative application of transfer learning in crop type mapping, the methodology proposed in this study effectively addressed the problem of missing labels in target domains and alleviated the negative impact of domain shift.

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Crop type mapping; Unsupervised domain adaptation; Time-series imagery; Transfer learning

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Introduction

Population growth and climate change pose considerable threats to food security, and there is an increasing demand for timely and reliable knowledge about crop geographical distribution to develop yield prediction models, generate agricultural statistics, and support food-security decision making (Wardlow and Egbert, 2008, Zhu et al., 2017). Traditional methods of obtaining crop type information involve field surveys and census, which require sophisticated survey infrastructure and often introduce time delays and subjectivity (Kuenzer and Knauer, 2013, Pott et al., 2021). Inter-annual variation of crop types implies frequent agricultural surveys, making large-area crop type mapping a more challenging task.

Satellite remote sensing, as an effective earth observation technique, provides a promising data source for crop type mapping. Coarse resolution imagery, such as from the Advanced Very High Resolution Radiometer(AVHRR) and the Moderate-resolution Imaging Spectroradiometer (MODIS), has been utilized to generate cropland extent at broad scales. Particularly, each crop has its unique phenology, and time series imagery allows the detection of vegetation phenological characteristics and thus better determination of crop types (Hao et al., 2020). However, coarse spatial resolution limits its applicability to croplands characterized by small patches and high spatial complexity. Landsat satellite imagery has a relatively high spatial resolution, making it easier to capture details of crop distribution. However, the 16-day repeat cycle, coupled with cloud contamination, may result in reduced data availability during critical periods of crop growth (Jönsson et al., 2018). Sentinel-2 images have demonstrated great potential for crop type mapping, which can be attributed to the finer spectral, spatial and temporal resolutions (Forkuor et al., 2018). Especially, Sentinel-2 incorporates three red-edge spectral bands centered at 705, 740, and 783 nm, the superiority of which has been emphasized by studies related to the detection of biophysical parameters such as green leaf area index (Delegido et al., 2011), Chlorophyll (Bramich et al., 2021) and nitrogen content (Clevers and Gitelson, 2013). Additionally, multi-temporal Sentinel-2 data have alleviated the difficulty in determining the optimal crop identification period (Vuolo et al., 2018), and effectively improved the accuracy of cropland extent identification (Watkins and Van Niekerk, 2019) and vegetation classification (Wakulińska and Marcinkowska-Ochtyra, 2020).

The construction of the relationship between spectral properties and crop types is the core process of remote sensing-based crop type mapping, and the performance of the algorithm that implements this process directly affects the final mapping accuracy. In this context, various machine learning methods like support vector machines (SVM) and random forest (RF) have been applied for crop classification (Saini and Ghosh, 2018). Recently, deep learning methods characterized by deeper and more complex structures have brought revolutionary advances in various research fields including agricultural applications (Feng et al., 2021b, Ma et al., 2021). Research on crop type mapping particularly benefit from deep learning because of its

ability to perform automatic high-level representations from data (Kussul et al., 2017). For example, long short-term memory (LSTM) network was employed to extract temporal characteristics from a sequence of Sentinel-2 observations to enhance the accuracy of crop classification (Rußwurm and Korner, 2017). Another study showed that convolution neural network (CNN) outperformed XGBoost, RF, and SVM, with an overall accuracy of 85.54% in summer crop classification (Zhong et al., 2019). Despite the success of these models, they require extensive agricultural ground truth to generate predictive function mapping from data to labels, and ground samples are usually collected through field surveys which are labor-intensive and time-consuming. Especially, it is difficult to conduct large-scale field surveys and to make findings public in many parts of the world due to economic, administrative, or other reasons.

Transfer learning is a machine learning method that uses knowledge gained from a previous task to enhance the learning of a related task. In the last few years, it has been introduced into remote sensing-based crop type mapping to address insufficient ground samples, thereby reducing labeling efforts (Hao et al., 2020, Nowakowski et al., 2021). Directly applying the models established in regions with abundant samples to other regions has been considered as a simple and effective transfer method. The continental United States (U.S.) is generally regarded as a sample-rich region due to the publicly available and widely used Cropland Data Layer (CDL). For example, high confidence pixels in the CDL and corresponding satellite based Normalized Difference Vegetation Index(NDVI) time series were selected to train the RF classification models, which were then applied to identify crop types in three target regions (Hao et al., 2020). In addition to cross-regional transfer, inter-annual transfer is also observed in crop type mapping related studies. For example, RF classifiers developed using the crop reference samples and Sentinel-1/2 data obtained in 2020 were successfully applied in 2019 to discriminate crop types in Hetao Irrigation District of China, with an overall accuracy of 0.92 (Hu et al., 2022). The intuition behind direct transfer is that crops of the same type have similar temporal growth patterns. However, due to meteorology-induced phenological and crop calendar differences, spectral information for crops of the same type may vary across regions and years. The difference between source and target data distribution is known as domain shift, which can explain the performance degradation in the application of the model from the source domain to the target domain (Gadiraju and Vatsavai, 2020). Since model building does not involve samples from the target domain, classification accuracy is more susceptible to domain shift when the fitted model is directly transferred and applied to a new region or year.

Fine-tuning is a common transfer learning technique to address the issue of domain shift. It uses a few samples from the target region to adjust the whole or part of the pre-trained model to adapt it to a new task. For example, fine-tuning technique was adopted to tune the parameters of the pre-trained VGG16 and GoogLeNet, and it was demonstrated to be effective in improving the accuracy of crop classification (Nowakowski et al., 2021). However, overfitting can occur during fine-tuning process, especially when only a small dataset is available in the target region. Therefore, data augmentation techniques have been used to expand the size of the data by adding modified or synthetic versions of the existing data. For example, six pre-trained CNNs were employed to identify crop diseases, and a data augmentation technique was utilized to improve generalization and prevent overfitting (Aravind and Raja, 2020). Compared with the aforementioned supervised methods, unsupervised domain

adaptation (UDA) methods can transfer knowledge learned from a source domain with a large number of labeled training samples to a target domain with only unlabeled data. As a UDA framework, a deep adaptation network (DAN) was proposed by Long et al. (2015), which can alleviate the negative impact of domain shift by aligning feature distributions between source and target domains. Given the effectiveness of DAN in transferable representation learning, its potential in crop type mapping deserves to be further explored.

In this study, we developed a deep adaptation crop classification network (DACCN) based on DAN framework to address the issue of missing labels of target regions and years during crop type mapping. Specifically, four states in the U.S. Corn Belt and two provinces in northeastern China were used as study areas, where samples used for model building and accuracy evaluation were collected based on time-series Sentinel-2 imagery and reference maps in 2018 and 2019. Three experiments were then designed to verify the effect of model transfer in space, time, and space–time respectively. In each experiment, the proposed DACCN was compared to two machine learning methods, RF and SVM, and a deep learning method, deep crop classification network (DCCN), which is structurally similar to DACCN but has no domain adaptation mechanism. Finally, the established models were used for crop type mapping in multiple regions, and the superiority of DACCN was further interpreted.

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Study areas

The U.S. and China are the leading agricultural producers in the world, and both have a significant impact on global food security. Therefore, this study selected four states in the U.S. (Fig. 1a) and two provinces in China (Fig. 1b) as the experimental areas. The four U.S. states are Minnesota (MN), Nebraska (NE), Wisconsin (WI), and Iowa (IA). They are all located in the Corn Belt, an agriculturally productive region that benefits from fertile soils, level land, and temperate climate. In 2018 …

Deep adaptation crop classification network (DACCN)

Traditional deep neural network (DNN) associates input samples $x \in X$ with labels $y \in Y$, where X and Y are input feature space and output space, respectively. Here in crop classification, x are spectral bands and VIs and y are crop type labels. Given a source domain $\mathscr{D}_s = \{(x_i^s, y_i^s)\}_{i=1}^{n_s}$ with n_s labeled samples and a target domain of $\mathscr{D}_t = \{(x_i^t, y_i^t)\}_{i=1}^{n_t}$ with n_t unlabeled samples were characterized by probability distributions p and q respectively. Generally, p is usually different from q due to …

Spectral characteristics of crops

To clarify the similarities and differences of the spectral characteristics of crops in each region and year, we presented the NDVI temporal profiles of corn, soybean, and "others" in Fig. 4. The mean profiles of "others" are significantly different from those of corn and soybean, and the standard deviation (std) of "others" is much larger at each time point. This is mainly because "others" includes various land use types, each with their own spectral characteristics. The NDVI profiles of corn …

Advantages of DACCN

Most existing crop type mapping related studies rely heavily on a large number of local labeled data for modeling and making predictions (Feng et al., 2019, Prins and Van Niekerk, 2021), and thus tedious and costly sample collection needs to be carried out extensively and frequently. The sharing of collected labeled samples is an effective way to address the dilemma of ground truth sampling. However, due to the differences in climate conditions across regions and years, trained crop …

Conclusions

In this paper, we proposed an UDA model DACCN for crop type mapping. Compared with traditional deep learning models that heavily rely on local labeled samples, DACCN is able to solve the label missing problem by learning knowledge from labeled samples in other domains and adaptively transferring the knowledge to the target domains. Three experiments were designed to evaluate the transferability of the proposed model across space, time, and space–time, respectively. It was found that DACCN had …

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. …

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